

Yi Xia (Sean)

PhD Candidate in Computer Vision & Human Activity Recognition · Auckland University of Technology

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70+
CITATIONS

3
H-INDEX

Q1
PAT. RECOG.

RESEARCH STATEMENT

I am a final-year PhD candidate at the **Auckland University of Technology (AUT)**, working at the intersection of **deep learning, computer vision, and graph neural networks**. My doctoral research designs *frequency-aware spatio-temporal topology learning* for skeleton-based human activity recognition (HAR), with a recent first-author paper accepted at *Pattern Recognition* (Q1, IF = 7.6). My earlier master's research on real-time object detection and tracking in agricultural video — the KiwiDetector / KiwiTracker line of work — has accumulated **50+ citations on a single paper**. I am building a research programme around *practical, interpretable, and computationally efficient* deep models for video and structured time-series data. Four years of parallel industry AI engineering at TOHU MEDIA in Auckland reinforce a deliberate stance: research that ships.

RESEARCH INTERESTS

- Skeleton-based human activity recognition
- Real-time object detection & tracking
- Frequency-domain representation learning
- Generative AI & agentic systems for vision workflows
- Spatio-temporal graph convolutional networks
- Video analytics for agriculture & media
- Attention & Transformer architectures for vision
- Practical, deployable deep-learning systems

EDUCATION

Doctor of Philosophy (PhD), Computer & Information Sciences — *Auckland University of Technology* Oct 2023 – Oct 2026 (expected)

Faculty of Design and Creative Technologies, School of Engineering, Computer & Mathematical Sciences. Member, Institute of Robotics & Vision (IoRV).
Provisional thesis: *Deep Learning for Skeleton-Based Human Activity Recognition*.
Supervisors: **Dr Raymond Lutui** (primary, AUT) · **Assoc. Prof. Sira Yongchareon** (AUT).
Funded by the **AUT PhD Full Scholarship** (tuition + living stipend, Oct 2023 – Oct 2026).

Master of Computer & Information Sciences (Computer Vision) — *Auckland University of Technology* 2021 – 2023

Research master. Thesis: *Kiwifruit Detection and Tracking from a Deep-Learning Perspective Using Digital Videos*.
Supervisors: Dr Minh Nguyen (primary) · Prof. Wei Qi Yan.

● PUBLICATIONS

Reverse chronological. Names in **bold** indicate first author. Citation counts as of April 2026; live counts on Google Scholar.

JOURNAL ARTICLES

- 01 **Xia, Y.**, Yongchareon, S., Lutui, R., & Sheng, Q. Z. (2026). Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition. *Pattern Recognition*.

Q1 · IF = 7.6

<https://doi.org/10.1016/j.patcog.2026.113146>

BOOK CHAPTERS

- 01 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2025). An Improved YOLO Model for Kiwifruit Detection. In *Optimization, Machine Learning, and Fuzzy Logic* (Advances in Computational Intelligence and Robotics series), IGI Global.

<https://doi.org/10.4018/979-8-3693-7352-1.ch008>

REFEREED CONFERENCE PAPERS

- 01 **Xia, Y.**, Yongchareon, S., Lutui, R., & Sheng, Q. Z. (2025). Dynamic Self-Attention Gated Spatial-Temporal Graph Convolutional Network for Skeleton-Based Human Activity Recognition. In *Proceedings of the 31st International Conference on Neural Information Processing (ICONIP 2024)*, Springer LNCS.

https://doi.org/10.1007/978-981-96-6588-4_14

- 02 **Xia, Y.**, Nguyen, M., Lutui, R., & Yan, W. Q. (2023). Multiscale Kiwifruit Detection from Digital Images. In *Proceedings of the Pacific-Rim Symposium on Image and Video Technology (PSIVT 2023)*, Springer. 7 cites

https://doi.org/10.1007/978-981-97-0376-0_7

- 03 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2023). Kiwifruit Counting Using KiwiDetector and KiwiTracker. In *Proceedings of the SAI Intelligent Systems Conference 2023*, Springer Lecture Notes in Networks and Systems. 11 cites

https://doi.org/10.1007/978-3-031-47724-9_41

- 04 **Xia, Y.**, Nguyen, M., & Yan, W. Q. (2022). A Real-Time Kiwifruit Detection Based on Improved YOLOv7. In *Proceedings of the 37th International Conference on Image and Vision Computing New Zealand (IVCNZ 2022)*, Springer LNCS. 50 cites

https://doi.org/10.1007/978-3-031-25825-1_4

THESES

- 01 **Xia, Y.** (2023). *Kiwifruit Detection and Tracking from a Deep-Learning Perspective Using Digital Videos*. Master's thesis, Auckland University of Technology.

● CONFERENCE PRESENTATIONS

- "Dynamic Self-Attention Gated Spatial-Temporal GCN for HAR," oral presentation, *ICONIP 2024*.
- "Multiscale Kiwifruit Detection from Digital Images," oral presentation, *PSIVT 2023*, Auckland, November 2023.
- "Kiwifruit Counting Using KiwiDetector and KiwiTracker," presentation, *SAI Intelligent Systems Conference*, 2023.

- "A Real-Time Kiwifruit Detection Based on Improved YOLOv7," oral presentation, *IVCNZ 2022*, Auckland, October 2022.

● RESEARCH EXPERIENCE

Doctoral Researcher — *Institute of Robotics & Vision (IoRV), AUT*

Oct 2023 – present

- Active member of the IoRV research group, contributing to its programme in deep learning, computer vision, and skeleton-based human activity recognition.
- Designed and implemented graph-convolutional architectures (FATL-GCN, Dynamic Self-Attention Gated ST-GCN) for skeleton-based HAR; trained and evaluated on NTU RGB+D 60/120 and related benchmarks.
- Published in *Pattern Recognition* (2026, Q1) and *ICONIP* (2024); two further first-author papers in preparation.

Research Assistant — *School of Engineering, Computer & Mathematical Sciences, AUT*

Nov 2022 – Jul 2023 (multiple appointments)

- AUT Summer Research Assistant (Nov 2022 – Feb 2023) under the AUT Summer RA Scholarship.
- Research Assistant (Jun – Jul 2023) under the SECMS Research Assistant Fund.
- Worked on the kiwifruit detection / counting research line, co-developing KiwiDetector and KiwiTracker; contributed to three peer-reviewed publications during these appointments.

● INDUSTRY EXPERIENCE

Artificial Intelligence Developer — *TOHU MEDIA · Taāmakī Makaurau / Auckland*

Feb 2022 – present

Māori-owned media-production company. Full-time since June 2025; previously part-time alongside MSc and PhD studies.

- Responsible for the company's AI capability end-to-end. Designed and shipped **three production models — object detection, OCR and action recognition** — replacing previously manual transcription work on client video footage with automated pipelines (specific deliverables under NDA).
- Engineer for the messy edge cases of real footage that off-the-shelf CV models stumble on: variable cinematography and framing, dense overlay text in non-standard layouts, archival material with degraded contrast, and long-form takes where naive per-frame inference doesn't fit a single-workstation GPU budget.
- Own the full ML lifecycle: dataset scope and labelling spec, training and evaluation in PyTorch, **Docker-packaged inference deployed to the company's cloud GPU servers**, and iteration with editors and the creative director on what "useful output" looks like inside an actual editing workflow. All models are trained and served on company-controlled infrastructure, preserving full data sovereignty over client footage.
- Demonstrates ability to translate research methods into production constraints and to communicate technical work with non-academic, creative-industry stakeholders.

Tech Service (part-time) — *PB Tech, Auckland*

Apr – Dec 2023

- Handled customer returns and exchanges, first-line hardware diagnostics and repairs, and custom-order PC assembly for walk-in customers.

• TEACHING EXPERIENCE

Teaching Assistant — *School of Engineering, Computer & Mathematical Sciences, AUT*

2022 – present

- TA on multiple undergraduate computer-science papers (programming, computer vision, AI).
- Lab leadership, marking, consultation hours, and mentoring of students on assignments and final-year projects.

Co-Founder — *Baiyin Pinuo Training School, China*

2019 – 2021

- Co-founded a private supplementary-education school; designed curriculum and led day-to-day academic operations. Divested in 2021 to relocate to New Zealand for postgraduate study; the school remains in operation under my co-founder.

Junior Lecturer, Department of Information Engineering — *Baiyin Vocational & Technology College, China*

Aug 2015 – May 2019

- Taught Computer Fundamentals, Java programming, IoT applications and RFID technologies to ~120 students per year over four academic years.
- Course design, lecture delivery, lab supervision, examinations and pastoral care.

• AWARDS & SCHOLARSHIPS

- **AUT PhD Full Scholarship** — tuition + living stipend (Oct 2023 – Oct 2026).
- **SECMS Research Assistant Fund** — May 2023, AUT.
- **AUT Summer Research Assistant Scholarship** — December 2022.

• PROFESSIONAL SERVICE & MEMBERSHIPS

PEER REVIEW

- Reviewer, *ACM Transactions on Multimedia Computing, Communications, and Applications* (TOMM), 2024.
- Reviewer, *International Agrophysics*, 2025 and 2026.
- Reviewer, *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications* (PIMRC), 2026.

MEMBERSHIPS

- Member, AUT Institute of Robotics & Vision (IoRV), 2022 – present.

• TECHNICAL SKILLS

Programming Python (expert · 8+ years) · Java (taught for 4 years).

Deep-learning frameworks PyTorch (primary) · TensorFlow / Keras · Hugging Face.

CV & scientific computing OpenCV · scikit-learn · NumPy · pandas · SciPy · Matplotlib.

Methods CNNs · GCNs · Transformers / attention · object detection (YOLO family) · object tracking · action recognition · OCR · semantic / instance segmentation · skeleton-based HAR · frequency-domain analysis.

Generative AI & LLM Claude / GPT API integration · agentic workflows · agent skills · tool-use & function calling · prompt engineering.

Deployment Docker · containerised model inference on cloud-hosted GPU servers.

Tooling Git / GitHub · Linux · CUDA · Jupyter.

● **LANGUAGES**

Mandarin Chinese <i>Native</i>	English <i>Professional working proficiency</i>
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● **REFEREES**

Available on request. Referees include the doctoral supervisory team (**Dr Raymond Lutui**, primary; **Assoc. Prof. Sira Yongchareon**, second supervisor — both AUT) and an industry referee from TOHU MEDIA.